



explain this in detail

Yes. This recording is **very important because it explains exactly how your supervisor wants the pricing MAB to be redesigned**. I'll explain it slowly and connect all the parts.

1. The main problem being discussed

Your current model already has:

- demand prediction
- supply information
- vehicle routing/repositioning probabilities
- destination information
- a Multi-Armed Bandit (MAB) that selects pricing factors
- customer acceptance/rejection
- dispatch and revenue calculation

But your supervisor is asking:

What information should the MAB use to select a pricing factor, and how should customer acceptance realistically depend on the offered price?

The entire meeting revolves around this question.

2. First: What is the "drop-off probability" or destination popularity?

Your supervisor says that the **drop-off probability is not simply an ID**.

Suppose you have destinations:

- Grid A
- Grid B
- Grid C

From historical ride data, you can calculate:

$$P(\text{Destination} = A)$$

$$P(\text{Destination} = B)$$

$$P(\text{Destination} = C)$$

For example:

Destination	Historical requests	Popularity
A	500	0.50
B	300	0.30
C	200	0.20

This probability represents:

How popular is this destination based on historical demand?

Your supervisor calls this a kind of **global popularity of the destination**.

Why not just give the destination ID?

Suppose:

- Destination ID = 25

The number **25 itself has no meaningful information** to the MAB.

But these are meaningful:

- destination popularity = 0.45
- distance = 8 km
- travel characteristics
- time of day
- weather

Therefore, instead of simply saying:

$$\text{Destination ID} = 25$$

you provide useful destination attributes.

3. Why does destination popularity matter for pricing?

Imagine two requests.

Request 1

- Destination: very popular

- Distance: 3 km
- Weather: normal
- High expected future demand

Request 2

- Destination: unpopular
- Distance: 3 km
- Weather: normal
- Low expected future demand

A driver dropping off in a popular destination may have a better chance of getting another ride.

Therefore, the trip can have a different value to the platform.

So your MAB can learn:

$$\text{Pricing Factor} = f(\text{Destination Popularity, Distance, Time, Weather, ...})$$

This is why destination popularity is an important attribute.

4. Supply information in your pricing model

This part is extremely important.

Your supervisor defines the **supply available in a grid** as:

$$S_{g,t} = \text{Idle Vehicles} + \text{Incoming Vehicles}$$

Idle vehicles

These are vehicles already waiting in grid g .

For example:

$$V_{g,t}^{idle} = 10$$

Incoming vehicles

These are vehicles that are currently completing trips and will arrive in this grid during the upcoming slot.

For example:

$$V_{g,t}^{incoming} = 5$$

Therefore:

$$S_{g,t} = 10 + 5 = 15$$

The incoming vehicles are not merely predicted randomly.

Since the dispatch system knows:

- when a trip starts
- origin
- destination
- expected travel duration

it can estimate in which future slot the vehicle will finish its trip.

Therefore, the model can maintain a counter of vehicles expected to arrive in each grid. This is exactly what the supervisor describes as vehicles waiting from the previous slot plus vehicles finishing trips and dropping off in that grid.

5. Predicted demand

The MAB also receives the predicted demand for the upcoming slot.

For example:

$$\hat{D}_{g,t+1} = 20$$

This tells the pricing model:

"How much demand is expected soon?"

So the MAB can make decisions using the relationship between:

Supply and Demand

For example:

Low supply + high demand

Potentially higher pricing factor.

High supply + low demand

Potentially lower pricing factor.

6. Routing/repositioning probability vector

Your supervisor also wants to provide information about:

$$P_{routing}$$

This represents how vehicles are likely to reposition.

Conceptually:

$$P_{g \rightarrow d}$$

represents the probability that a vehicle from grid g will move toward destination d .

The key point in the meeting is that you should not necessarily provide the **entire large routing vector as one giant input**.

Instead, you consider one destination/request at a time and use the relevant information for that destination.

7. So what exactly becomes the MAB input?

This is the central architecture.

For a request going from grid g to destination d , the MAB can receive:

$$X_{g,d,t} = [S_{g,t}, \hat{D}_{g,t+1}, P_{route}, \text{Destination Popularity, Distance, Base Price, Time, Weather}]$$

Let's break it down.

A. Supply

$$S_{g,t} = \text{Idle Vehicles} + \text{Incoming Drop-offs}$$

B. Predicted demand

$$\hat{D}_{g,t+1}$$

C. Routing probability

The repositioning/routing information associated with the destination.

D. Destination popularity

$$P(d)$$

How popular is this destination historically?

E. Distance to destination

$$Dist(o, d)$$

F. Base price

A reference price.

G. Time of day

For example:

- morning
- afternoon
- evening peak
- night

H. Weather

For example:

- normal
- rain
- heavy rain
- extreme weather

This is very close to the architecture your supervisor explicitly lists in the meeting.

8. What does the MAB produce?

The MAB produces a pricing factor:

$$\alpha$$

For example:

$$\alpha \in \{0.85, 0.90, \dots, 1.15\}$$

Then:

$$P_{\text{dispatch}} = P_{\text{base}} \times \alpha$$

Example:

$$P_{\text{base}} = ₹100$$

and:

$$\alpha = 1.20$$

Therefore:

$$P_{\text{dispatch}} = ₹120$$

This is what the customer actually sees.

The customer **does not see** $\alpha = 1.20$.

The customer sees:

₹120 for this ride.

This distinction is very important in your supervisor's discussion.

9. The biggest question: How does the customer decide to accept or reject?

Your current implementation probably does something similar to:

$$P(\text{Accept}) = \text{some function of price}$$

Then:

$$Y \sim \text{Bernoulli}(P(\text{Accept}))$$

where:

$$Y = 1$$

means accepted and:

$$Y = 0$$

means rejected.

Your supervisor says:

The acceptance probability should not be a fixed flat probability.

Why?

Because a customer's reaction to price changes depending on the context.

10. Customer sensitivity is context-dependent

Suppose the offered price is ₹200.

The customer response may differ.

Scenario 1: Pleasant weather

$$P(\text{Accept}) = 0.40$$

The customer may think:

"This is too expensive. I can wait."

Scenario 2: Heavy rain

$$P(\text{Accept}) = 0.80$$

The same ₹200 might be accepted because transportation is urgently needed.

Therefore:

$$\text{Price Sensitivity} = f(\text{Time, Weather, Distance, ...})$$

Your supervisor says customers may be **less sensitive to price in harsh conditions** and **more sensitive in convenient conditions**.

11. The supervisor gives two possible approaches

Approach 1: Directly provide customer sensitivity

First, study customer behavior from data.

For example, estimate:

$$\beta_{\text{weather}}$$

$$\beta_{\text{time}}$$

$$\beta_{\text{distance}}$$

Then provide the estimated customer sensitivity directly to the MAB.

Conceptually:

$$X = [\text{Supply, Demand, Routing, Sensitivity}]$$

The MAB knows:

"Under this weather and time, customers are highly/less price-sensitive."

This approach is explicitly discussed in the recording.

12. Approach 2: Give the factors that influence sensitivity

Instead of directly calculating sensitivity first, give the MAB:

- time
- weather
- distance
- base price

The model receives the same contextual information that influences customer behavior.

Then:

MAB learns

how pricing should vary under different contexts.

Conceptually:

$$X = [\text{Supply, Demand, Routing, Time, Weather, Distance, Base Price}]$$

This is probably the simpler and more direct approach for your current implementation.

13. Why does the MAB need the same information that affects customer behavior?

This is a very important conceptual point.

Suppose the customer acceptance is actually determined by:

$$P(\text{Accept}) = f(\text{Price, Weather, Time})$$

But the MAB only sees:

[Supply, Demand]

Then the MAB is trying to learn customer behavior **without seeing the variables that cause the behavior to change**.

This makes learning harder.

Instead, give:

[Supply, Demand, Weather, Time, Price Information]

Then the MAB can learn:

"During heavy rain, customers tolerate a higher price."

or:

"During convenient weather, customers reject higher prices."

That is exactly the logic behind the supervisor saying that the MAB should at least receive the same information based on which customers react.

14. Customer acceptance function

Your supervisor refers to a sigmoid/logistic-type function.

Generally:

$$P(\text{Accept}) = \frac{1}{1 + \exp(-z)}$$

where z can contain:

$$z = \beta_0 + \beta_1(\text{Price Difference}) + \beta_2(\text{Weather}) + \beta_3(\text{Time}) + \beta_4(\text{Distance})$$

Therefore:

$$P(\text{Accept}) = \frac{1}{1 + \exp(-[\beta_0 + \beta_1 \Delta P + \beta_2 W + \beta_3 T + \beta_4 D])}$$

The customer then accepts or rejects probabilistically.

The supervisor is essentially saying that **the probability must depend on the context rather than being a constant.**

15. What should be used as the price variable?

This is another important point.

The customer sees:

$$P_{\text{dispatch}}$$

not:

$$\alpha$$

Therefore, the acceptance model should ideally compare:

$$P_{\text{dispatch}}$$

with:

$$P_{\text{base}}$$

For example:

$$P_{\text{base}} = ₹100$$

$$P_{\text{dispatch}} = ₹130$$

Then:

$$\Delta P = 130 - 100 = ₹30$$

or:

$$\text{Relative Price Increase} = \frac{130 - 100}{100} = 0.30$$

The customer can react to this 30% increase.

This is much more meaningful than directly reacting to:

$$\alpha = 1.3$$

16. Offline approach to estimate sensitivity

The supervisor then discusses an **offline approach**.

This means:

Before simulation:

Use historical data to study:

$$\text{Price} \leftrightarrow \text{Acceptance/Demand}$$

under different conditions.

For example:

$$P(\text{Accept}) = f(\text{Price})$$

for different:

- weather conditions
- times of day

- distances

You could estimate curves such as:

$$P(\text{Accept}) = f(\text{Price} \mid \text{Rain})$$

and:

$$P(\text{Accept}) = f(\text{Price} \mid \text{No Rain})$$

Then derive sensitivity from these curves.

The supervisor suggests fitting linear, exponential, or sigmoid-type functions depending on the data.

17. What is “price elasticity” here?

Price elasticity tells you:

How strongly does demand or acceptance change when price changes?

Conceptually:

$$E = \frac{\% \text{ change in demand}}{\% \text{ change in price}}$$

A high magnitude means customers are very price-sensitive.

Example:

Group A

$$E = -2$$

A 1% price increase causes roughly a 2% reduction in demand.

Group B

$$E = -0.5$$

Customers are less sensitive.

The supervisor wants this sensitivity potentially to vary according to:

$$E = f(\text{Weather}, \text{Time}, \text{Distance})$$

18. Another possible approach: sensitivity distribution

Instead of one fixed elasticity:

$$E = -1$$

you can model a distribution:

$$E \sim N(\mu_E, \sigma_E^2)$$

For example:

Normal weather

$$E \sim N(-1.5, 0.3^2)$$

Heavy rain

$$E \sim N(-0.7, 0.2^2)$$

This represents different types of customers.

Some customers may be:

- highly price-sensitive
- moderately price-sensitive
- less price-sensitive

The supervisor discusses estimating such sensitivity distributions offline, or learning them through a more advanced model, but treats online learning of the distribution as beyond the immediate scope.

19. What happens after customer acceptance?

Suppose:

$$P_{\text{dispatch}} = P_{\text{base}} \times \alpha$$

The customer accepts.

Then the dispatch process happens.

After that, the system calculates the outcome and reward.

For example:

$$R_t = \text{Revenue} - \text{Costs}$$

The customer response is part of the feedback.

The learning loop is:

Context $\rightarrow$ MAB chooses α



$$P_{\text{dispatch}} = P_{\text{base}} \times \alpha$$



Customer Accept/Reject



Revenue / Reward



Update LIN-UCB

Your supervisor specifically asks where the customer response enters the feedback loop and explains that it can influence the reward and updates to the linear models associated with the MAB arms.

20. The major contribution of your model

Your supervisor compares your proposed model with the existing reference approach.

A baseline approach may primarily learn:

Context $\rightarrow$ Pricing Factor

But your proposed model is intended to have a **richer understanding of both supply-side and demand-side behavior**.

Supply-side behavior

- idle vehicles
- incoming vehicles
- predicted demand
- routing/repositioning probabilities

Demand-side/customer behavior

- customer acceptance
- weather-dependent sensitivity

- time-dependent sensitivity
- distance effects
- base price versus offered price

The argument is:

Better modeling of both supply and customer behavior should enable better pricing decisions and therefore potentially improve revenue and operational performance.

The supervisor explicitly connects MAB performance with having a better understanding of supply-side and demand-side behavior.

21. How you can compare your model with the existing paper

This part is important for your PhD.

You should create different versions.

Baseline

Use the existing paper's inputs and approach.

Measure:

- total platform revenue
- average vehicle revenue
- completed trips
- repositioning cost

Your improved model

Add:

- routing/repositioning information
- destination popularity
- time of day
- weather
- distance
- base price/reference information
- customer sensitivity modeling

Then compare:

Baseline Revenue

versus:

Proposed Model Revenue

Also compare:

Repositioning Cost

Average Vehicle Revenue

Requests Served

The goal is to demonstrate:

Better context $\Rightarrow$ Better pricing decisions $\Rightarrow$ Better system performance

This comparison logic is described in the recording.

22. The complete architecture you should now understand

Input layer

Supply-side

$S_t = \text{Idle Vehicles} + \text{Incoming Drop-offs}$

$\hat{D}_{t+1} = \text{Predicted Demand}$

$P_{route} = \text{Routing Probability}$

Destination attributes

[Popularity, Distance]

Pricing/customer attributes

[Base Price, Time, Weather]

MAB pricing layer

For each request:

$$X_t = [S_t, \hat{D}_{t+1}, P_{route}, \text{Popularity}, D, P_{base}, T, W]$$

Then:

$$\text{LIN-UCB}(X_t) \rightarrow \alpha_t$$

Price calculation

$$P_{\text{dispatch},t} = P_{\text{base},t} \times \alpha_t$$

Customer behavior model

$$P(\text{Accept}) = f(P_{\text{dispatch}}, P_{\text{base}}, D, T, W, \text{Sensitivity})$$

Then:

Accept/Reject

Dispatch and reward

If accepted:

Dispatch $\rightarrow$ Served/Unserved

Then calculate:

$$\text{Reward} = \text{Revenue} - \text{Cost}$$

and update LIN-UCB.

23. The key change you need to make

The simplest way to understand your supervisor's recommendation is:

Your old idea

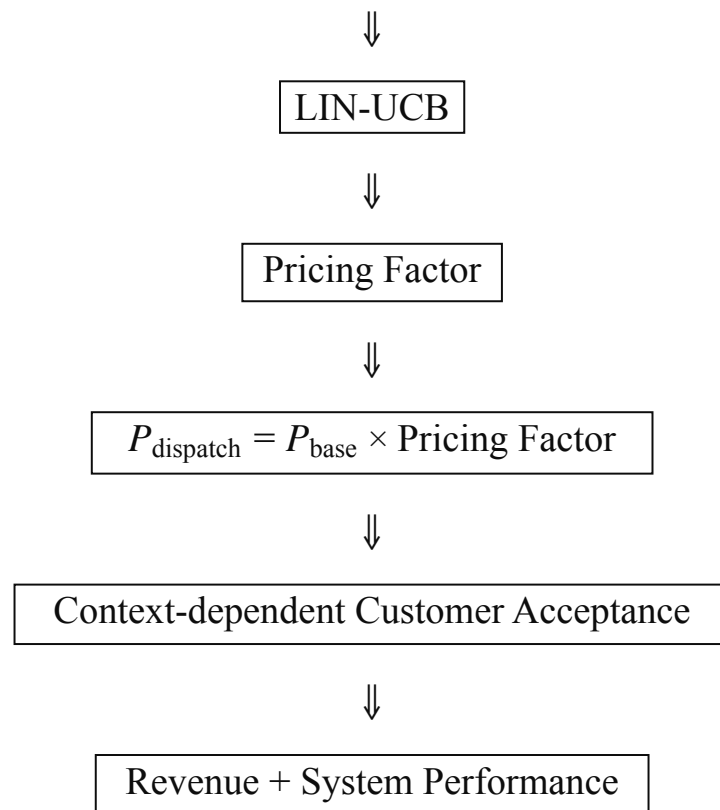
"Give the MAB some operational information and let it choose a pricing factor."

Your new idea

"Give the MAB detailed supply-side information, destination information, and the contextual variables that influence customer price sensitivity, then let it learn the appropriate pricing factor."

So your central model becomes:

Supply State + Demand Prediction + Routing Probability + Destination Pop
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My view of the most important message from this recording

The supervisor is not simply asking you to add more inputs randomly. He wants a logical connection between:

1. The context in which a price is chosen
2. The actual price offered
3. The context in which a customer decides to accept/reject
4. The feedback used by LIN-UCB to learn

Your model should therefore explain **both sides**:

- Why the platform chooses a price
- Why the customer accepts or rejects that price

That is the main conceptual foundation of the implementation described in this meeting.



AudioCutter Record-01...
PDF

[explain this in detail](#)

This third recording explains **four major things together**:

1. **How to calculate customer price sensitivity/elasticity from historical data**
2. **How to convert that elasticity into a customer accept/reject model**
3. **How to show convergence for both the FL model and the MAB model**
4. **What final system-performance results and ablation studies you should produce**

It also gives some important clarification about **charging** and **how you should position your overall research contribution**.

PART 1: CUSTOMER PRICE ELASTICITY

This is the first major topic.

Your supervisor says that the customer should not have one fixed sensitivity to price. Instead, customer price sensitivity should depend on contextual conditions such as:

- distance
- time of day
- weather

He uses **price elasticity** and **customer sensitivity** almost interchangeably in this discussion.

1.1 What is the basic idea?

You have a reference or base situation:

$$P_{\text{base}}$$

$$D_{\text{base}}$$

$$Dist_{\text{base}}$$

You then observe another situation:

$$P_i, \quad D_i, \quad Dist_i$$

Now calculate relative/percentage changes.

Price change

$$\Delta P_i = \frac{P_i - P_{\text{base}}}{P_{\text{base}}}$$

Demand change

$$\Delta D_i = \frac{D_i - D_{\text{base}}}{D_{\text{base}}}$$

Similarly, you can calculate a distance-related relative change:

$$\Delta \text{Dist}_i = \frac{\text{Dist}_i - \text{Dist}_{\text{base}}}{\text{Dist}_{\text{base}}}$$

The supervisor discusses constructing elasticity/sensitivity values using percentage changes with respect to reference values.

1.2 How does elasticity enter the equation?

The basic relationship discussed is conceptually:

$$\% \Delta P = \frac{1}{\epsilon} \% \Delta D$$

or equivalently:

$$\% \Delta D = \epsilon \% \Delta P$$

where:

$$\epsilon$$

is the price elasticity.

For each historical observation, you can obtain an elasticity-like ratio:

$$\epsilon_i = \frac{\% \Delta D_i}{\% \Delta P_i}$$

This means that historical data gives you **many elasticity values**, not just one.

For example:

$$[-0.8, -1.2, -1.0, -1.7, -0.9, \dots]$$

The negative sign normally indicates that demand decreases when price increases.

1.3 Why does the supervisor mention Euclidean norm?

Instead of simply calculating one ratio for every observation, the supervisor describes a least-squares-style approach.

Suppose you define:

$$A = \begin{bmatrix} \Delta P_1 \\ \Delta P_2 \\ \vdots \\ \Delta P_n \end{bmatrix}$$

and:

$$B = \begin{bmatrix} \Delta D_1 \\ \Delta D_2 \\ \vdots \\ \Delta D_n \end{bmatrix}$$

Then you want a single scalar elasticity x such that:

$$Ax \approx B$$

You find the x that minimizes:

$$\|Ax - B\|_2$$

This is what he means by:

"minimize the Euclidean norm."

The solution in this simple one-variable least-squares case is:

$$x = \frac{A^T B}{A^T A}$$

So, the important point is:

You can estimate one representative elasticity value from many historical observations by finding the best relationship between percentage changes.

PART 2: DO NOT USE ONE ELASTICITY FOR EVERYONE

This is the next important idea.

Suppose you calculate all historical elasticity values for:

- morning
- normal weather

You may obtain:

$$[-0.8, -1.0, -1.4, -0.9, -1.2]$$

Instead of using:

$$\epsilon = -1.06$$

for every customer, calculate:

$$\mu_{\epsilon} = \text{mean elasticity}$$

and:

$$\sigma_{\epsilon} = \text{standard deviation of elasticity}$$

Then customer sensitivity can be modeled as:

$$\epsilon \sim N_{truncated}(\mu_{\epsilon}, \sigma_{\epsilon}^2)$$

The supervisor explicitly suggests using the mean and standard deviation from historical elasticity values to create a **truncated normal distribution**.

2.1 Why a truncated normal distribution?

Because you probably do not want completely unrealistic elasticity values.

For example, suppose:

$$\epsilon \sim N(-1.0, 0.3^2)$$

The model can randomly generate:

- Customer 1: -0.8
- Customer 2: -1.1
- Customer 3: -1.3

So different customers have different sensitivity.

This makes the simulation more realistic.

2.2 Sensitivity changes with time and weather

You should not estimate one global distribution.

Instead:

$$\epsilon^{(t,w)} \sim N_{truncated}(\mu_{\epsilon}^{(t,w)}, (\sigma_{\epsilon}^{(t,w)})^2)$$

where:

- t = time of day
- w = weather condition

For example:

Morning + normal weather

$$\epsilon^{(morning,normal)} \sim N(-1.5, 0.3^2)$$

Evening + heavy rain

$$\epsilon^{(evening,rain)} \sim N(-0.6, 0.2^2)$$

The interpretation is that customers may have different price sensitivity under different conditions.

The recording clearly says that a separate sensitivity distribution can be calculated for each time–weather pair.

PART 3: HOW MANY HISTORICAL GROUPS DO YOU CREATE?

Suppose you have:

- 24 time slots
- 3 weather conditions

Then:

$$24 \times 3 = 72$$

different groups.

For each group, you collect the relevant historical data.

So:

$$D^{(t,w)}$$

is the historical data corresponding to:

$$(t, w)$$

For each group:

1. Select historical observations.
2. Determine the reference/base price.

3. Calculate relative changes.
4. Estimate elasticity values.
5. Calculate mean elasticity.
6. Calculate standard deviation.
7. Construct a truncated distribution.

This gives you:

72

different sensitivity distributions in this example.

PART 4: HOW THE CUSTOMER ACTUALLY ACCEPTS OR REJECTS

This is one of the most important implementation steps.

The workflow suggested by your supervisor is:

Step 1: Select the current context

For example:

- Time = 18:00–18:30
- Weather = heavy rain

Therefore use:

$$\epsilon^{(18:00, \text{rain})}$$

Step 2: Randomly generate a customer sensitivity

Sample:

$$\epsilon_c \sim N_{truncated}(\mu_{\epsilon}^{(t,w)}, (\sigma_{\epsilon}^{(t,w)})^2)$$

This represents the sensitivity of the current customer.

Step 3: Calculate the customer's maximum acceptable price

Your supervisor describes constructing an upper price limit using:

- reference price
- reference/base distance

- actual trip distance
- sampled elasticity

Conceptually:

$$P_{\max} = P_{\text{ref}} + P_{\text{ref}} \left[\frac{1}{\epsilon_d} \frac{Dist - Dist_{\text{ref}}}{Dist_{\text{ref}}} \right]$$

The exact sign and final implementation should be checked carefully against the intended elasticity convention, but the core idea is clear:

The customer's sensitivity and trip characteristics are used to generate the maximum price the customer is willing to pay.

The supervisor describes exactly this process: sample a sensitivity, calculate an upper price limit, then compare the operator's offered price against that limit.

PART 5: ACCEPT/REJECT RULE

The MAB selects:

$$\alpha$$

Then the platform calculates:

$$P_{\text{offer}} = P_{\text{base}} \times \alpha$$

The customer has:

$$P_{\max}$$

Then:

Accept if $P_{\text{offer}} \leq P_{\max}$

Otherwise:

Reject

So your customer model is:

Historical Data → Elasticity Distribution → Sample Customer Sensitivity → $P_{\max}$

Then:

$$P_{\text{offer}} = P_{\text{base}} \times \alpha$$

and finally:

$$P_{\text{offer}} \leq P_{\text{max}} \Rightarrow \text{Accept}$$

This is the **offline customer behavior model** your supervisor is describing.

PART 6: TWO POSSIBLE WAYS TO USE SENSITIVITY IN THE MAB

Your supervisor again gives you two alternatives.

Option A: Explicitly give sensitivity information

Input to the MAB:

$$X = [\text{Supply, Demand, Routing, Sensitivity Information}]$$

The MAB receives the sensitivity distribution or sensitivity-related values directly.

For example:

$$[\mu_{\epsilon}, \sigma_{\epsilon}]$$

Option B: Give the contextual variables instead

Do not explicitly give sensitivity.

Give:

$$[\text{Time, Weather, Distance, Base Price}]$$

The MAB learns how those conditions affect pricing decisions.

The supervisor says both approaches are possible.

PART 7: NOW YOUR SUPERVISOR MOVES TO RESULTS

You showed results like:

- mean requests
- accepted requests
- rejected requests

The supervisor says:

Those results tell us system performance, but they do not prove that the MAB actually learned.

This distinction is critical.

PART 8: HOW TO SHOW MAB CONVERGENCE

You need to show whether:

$$\text{MAB learning} \rightarrow \text{Convergence}$$

The supervisor suggests two possible ways.

Method 1: Objective/reward convergence

For every episode:

$$J_1, J_2, \dots, J_N$$

where J could be:

- reward
- revenue
- utility

Then calculate and plot:

$$J_k$$

over episodes.

If it stabilizes, the MAB may have converged.

Method 2: Pricing factor convergence

For each episode, analyse:

$$\alpha$$

You want to see whether the pricing policy becomes stable.

The supervisor specifically says individual request-level selection is not what he wants to see. He is more interested in **episode-level behavior and convergence**.

PART 9: FEDERATED LEARNING CONVERGENCE

This is another separate learning process.

You have:

1. FL learning routing/repositioning behavior
2. MAB learning pricing

Therefore, you need results for both.

Your supervisor says there are two possible ways to analyse FL convergence.

Option A: Local model MSE

Each vehicle trains its local model.

For vehicle v :

$$MSE_v^{(r)}$$

where r is the FL round.

You can observe:

$$MSE^{(1)} \rightarrow MSE^{(2)} \rightarrow MSE^{(3)} \rightarrow \dots$$

If it decreases and stabilizes:

FL model is learning

You could:

- plot selected representative vehicles
 - calculate mean MSE across vehicles
 - analyse global/average MSE
-

Option B: Global model parameter convergence

This is the method your supervisor explains in more detail.

Let the global model after FL round r be:

$$\theta^{(r)}$$

Then calculate:

$$\Delta\theta^{(r)} = \theta^{(r)} - \theta^{(r-1)}$$

Then compute:

$$C_r = \left\| \theta^{(r)} - \theta^{(r-1)} \right\|_2$$

This is the Euclidean norm of the change in global model parameters.

If:

$$C_r \rightarrow 0$$

then the global model is stabilizing.

Example:

Round	Parameter change
1	5.20
10	1.40
20	0.30
30	0.05
40	0.009

You can set a threshold:

$$C_r < 0.01$$

and say the global model has effectively converged.

This is directly described in the recording.

PART 10: YOU HAVE TWO LEARNING ALGORITHMS

This is an important result-design principle.

Your supervisor says:

FL Model

and:

MAB Model

are learning simultaneously.

Therefore, you should report both:

FL performance

- MSE
- parameter convergence
- routing probability prediction performance

MAB performance

- reward/revenue convergence
- pricing factor convergence

You should **not show only final system results**.

You must show:

1. Did FL learn?
2. Did MAB learn?
3. Did the complete system perform well?

PART 11: SYSTEM PERFORMANCE

Now, after algorithm convergence, measure what the agents achieved.

For the platform/operator

Calculate:

Mean System Revenue

For vehicles

Calculate:

Mean Vehicle Utility

or:

Mean Vehicle Revenue

Also look at:

Mean Total Revenue Per Day

The main question is:

Did your model help the platform earn more?

and:

Did your vehicle-control mechanism help vehicles achieve better utility/revenue?

The supervisor explicitly asks for these agent-level performance measures.

PART 12: WHY SOME RESULTS ARE CURVES AND OTHERS ARE BARS

This is very important for your final paper.

Curves

Use curves for:

Learning over time

- FL convergence
- MAB convergence
- reward over episodes
- parameter changes over rounds

Because there is a sequence:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow \dots \rightarrow N$$

Bar charts

Use bar charts for comparison.

For example:

Model	Revenue
Without weather	100
With weather	120
Without routing	95
Proposed model	135

These are single final performance values.

Your supervisor says when you vary conditions or compare methods, you will obtain numbers that are suitable for bar-chart comparisons.

PART 13: ABLATION STUDY

This is one of the most important parts for your PhD.

An ablation study asks:

What happens when one component of the proposed model is removed?

Your supervisor suggests comparisons such as:

Without weather

Model – Weather

Without destination popularity

Model – Popularity

Without routing probability

Model – Routing

Without supply-side behavior

Then compare:

- total revenue
- average vehicle revenue
- dispatch success
- number of served requests

This demonstrates the contribution of each component.

For example:

Model	Revenue	Served Requests
Full model	1000	900
Without weather	920	850
Without popularity	880	830
Without routing	800	750

Then you can say:

Routing information contributes X% improvement.

This is exactly why your supervisor discusses removing weather, popularity, and routing information and comparing performance.

PART 14: CHARGING RESULTS

Now the discussion moves to charging.

Your supervisor says the results should also explain:

How did vehicles benefit from charging?

Potential metrics include:

1. Mean number of vehicles able to charge

$$\overline{N}_{charge}$$

2. Battery degradation cost

$$\overline{C}_{degradation}$$

3. Utility cost / charging-related utility

Your supervisor is saying that you should not only implement charging—you should measure its impact.

PART 15: YOUR MODEL IS NOT ONLY FOR EVs

This is another important conceptual clarification.

Your model contains charging conditions.

But your supervisor says you should not present the entire research as:

"A model only for electric vehicles."

Instead, the broader model is for vehicle dispatching/repositioning/pricing.

Then charging is an additional requirement for vehicles that need charging.

So the model can support a mixed fleet:

10% EVs

25% EVs

50% EVs

100% EVs

This is important because the model can be tested at different EV penetration levels.

PART 16: EV PENETRATION STUDY

You can perform:

EV Penetration	System Performance
10%	...
25%	...
50%	...
75%	...
100%	...

Then ask:

As more EVs enter the system, does charging infrastructure become a bottleneck?

Suppose performance falls at:

50% EV penetration

Then increase:

Charging Capacity

and test again.

For example:

Chargers	Served Requests
10	800
20	870
30	910

This allows you to demonstrate:

The framework can support EV fleets, and system performance improves when sufficient charging capacity is available.

PART 17: THE PAPER CONTRIBUTION

The last part discusses what should be highlighted about existing work.

Your supervisor is building the literature comparison around the limitations of previous models.

Existing works may:

- focus only on dispatching
- focus on charging for EVs
- focus on dispatching and repositioning
- consider dispatch pricing
- reposition vehicles only based on demand
- use demand-supply imbalance
- use incentives for repositioning

Your contribution is not merely:

"I have another dispatch algorithm."

The major point is that your work aims to combine and improve these decisions.

PART 18: THE BIGGEST EXISTING LIMITATIONS

Your supervisor specifically highlights these assumptions in prior work.

1. Vehicles always follow repositioning recommendations

Previous work may assume:

Recommended destination = Actual destination chosen by vehicle

But in reality:

$$P(\text{Vehicle follows recommendation}) < 1$$

Your model explicitly learns routing/repositioning probabilities.

2. No failure/uncertainty modeling

Previous work may assume decisions are perfectly followed.

Your model considers uncertainty in actual vehicle behavior.

3. Existing methods assume known vehicle preferences

Previous work may assume the platform already knows:

$$P(\text{Vehicle chooses destination})$$

But your work tries to **learn** this behavior.

4. Probabilities are not learned

Some approaches may use:

- arbitrary routing probabilities
- predefined probabilities

Your model instead uses:

Federated Learning to learn routing probabilities

5. Vehicles are assumed homogeneous

Many models assume:

$$\text{Vehicle}_1 = \text{Vehicle}_2 = \dots$$

But your work allows vehicle-level behavior/preferences to be represented more realistically.

These limitations are explicitly summarized at the end of the recording.

THE COMPLETE PICTURE

The whole model now looks like this:

Historical Data



Estimate Customer Elasticity by Time + Weather



$$\mu_{\epsilon}^{(t,w)}, \sigma_{\epsilon}^{(t,w)}$$

At simulation time:

Current Request



Supply-side information

Idle Vehicles + Incoming Vehicles

Demand-side information

Predicted Demand

FL-based vehicle behavior

Routing Probabilities

Request/customer information

Distance + Time + Weather + Destination Attributes



LIN-UCB / MAB



Pricing Factor



$$P_{\text{offer}} = P_{\text{base}} \times \alpha$$



Sample:

$$\epsilon_c \sim N_{\text{truncated}}(\mu_{\epsilon}^{(t,w)}, \sigma_{\epsilon}^{(t,w)})$$



Calculate:

$$P_{\max}$$

Then:

$$P_{\text{offer}} \leq P_{\max}$$

→ **Accept**

Otherwise:

→ **Reject**

Then your evaluation has three levels

Level 1: Did the algorithms learn?

FL

$$\|\theta^{(r)} - \theta^{(r-1)}\|_2$$

or MSE convergence.

MAB

- reward convergence
 - objective convergence
 - pricing-policy convergence
-

Level 2: Did the system perform better?

- total system revenue
 - vehicle revenue/utility
 - requests accepted
 - requests served
 - wait time
 - repositioning cost
 - charging performance
-

Level 3: Why did the model perform better?

Ablation studies:

- without weather
- without destination popularity
- without routing probabilities
- without supply-side information
- without other proposed components

This is the full logic your supervisor is asking you to demonstrate.

In one sentence

First, learn realistic customer sensitivity from historical data; second, use that to simulate customer response; third, prove both FL and MAB convergence; fourth, prove the proposed model improves system and vehicle performance; finally, use ablation studies to show which components create that improvement.